Introduction to Shell Scripting

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What is the Shell?

It is...

...a program

...a programming language

...how you interact with the operating system

A first shell script

first.sh

```
# Comments are prefixed with a # mark

echo "It is now" `date`
echo
who
echo
echo "My user name is" $(whoami) "and I am in
directory" $(pwd)
```

The output is:

```
rdickson@glooscap: ~/demo > chmod +x first.sh
rdickson@glooscap: ~/demo > ./first.sh
It is now Thu Oct 30 10:54:20 ADT 2008

skhan    pts/0        Oct 30 08:29 (140.184.24.201)
skhan    pts/3        Oct 30 08:47 (140.184.24.201)
vchevrie pts/5        Oct 30 09:42 (happy.phys.dal.ca)
rdickson pts/6        Oct 30 10:41 (rdickson.ucis.dal.ca

My user name is rdickson and I am in directory
/home/rdickson/demo
rdickson@glooscap: ~/demo >
```

A practical example:

Suppose you have
fifty jobs on the cluster,
some in *r* state,
some in *qw* state,
and some in *Eqw* state.

How do you delete your *Eqw* jobs efficiently?

qstat output

```
 $ qstat
job-ID  prior   name       user         state submit/start at     queue
-----------------------------------------------------------------------------
4320603 0.50000 qgenNettsZ jksmith      r     05/21/2010 15:22:58 long.q
4320606 0.50000 qgenNettsZ jksmith      r     05/21/2010 15:23:14 long.q
4320607 0.50000 qgenNettsZ jksmith      r     05/21/2010 15:23:14 long.q
4320604 0.50000 qgenNettsZ jksmith      r     05/21/2010 15:22:58 long.q
4320605 0.50000 qgenNettsZ jksmith      r     05/21/2010 15:22:58 long.q
4320600 0.50000 qgenNettsZ jksmith      r     05/21/2010 15:22:28 long.q
4319822 0.50000 qpostPro   jksmith      qw    05/21/2010 14:13:17
4319823 0.50000 qpostPro   jksmith      qw    05/21/2010 14:13:22
4319826 0.50000 qpostPro   jksmith      qw    05/21/2010 14:13:30
4319828 0.50000 qpostPro   jksmith      qw    05/21/2010 14:13:35
4319829 0.50000 qpostPro   jksmith      qw    05/21/2010 14:13:41
4319831 0.50000 qpostPro   jksmith      qw    05/21/2010 14:13:47
4319837 0.50000 qpostPro   jksmith      Eqw   05/21/2010 14:14:05
4319839 0.50000 qpostPro   jksmith      Eqw   05/21/2010 14:14:11
4319841 0.50000 qpostPro   jksmith      Eqw   05/21/2010 14:14:17
4319843 0.50000 qpostPro   jksmith      Eqw   05/21/2010 14:14:23
```

We need to 'qdel 4319837', 'qdel 4319839', ...

The answer

Suppose you have fifty jobs on the cluster, some in *r* state, some in *qw* state and some in *Eqw* state

How do you delete your *Eqw* jobs efficiently?

```
qstat > temp1

grep 'Eqw' temp1 > temp2

cut -c1-8 temp2 > temp3

sed 's/^/qdel /' temp3 > del.sh

source del.sh

rm temp1 temp2 temp3 del.sh
```

This script will do it ---
Let's examine the pieces

Eqw.sh

Find lines with *grep*

```
 $ qstat
job-ID  prior   name       user         state submit/start at     queue
-----------------------------------------------------------------------
4320605 0.50000 qgenNettsZ jksmith      r     05/21/2010 15:22:58 long.q
4320600 0.50000 qgenNettsZ jksmith      r     05/21/2010 15:22:28 long.q
4319829 0.50000 qpostPro   jksmith      qw    05/21/2010 14:13:41
4319831 0.50000 qpostPro   jksmith      qw    05/21/2010 14:13:47
4319837 0.50000 qpostPro   jksmith      Eqw   05/21/2010 14:14:05
4319839 0.50000 qpostPro   jksmith      Eqw   05/21/2010 14:14:11
4319841 0.50000 qpostPro   jksmith      Eqw   05/21/2010 14:14:17
4319843 0.50000 qpostPro   jksmith      Eqw   05/21/2010 14:14:23
```

grep 'Eqw' temp1

```
4319837 0.50000 qpostPro   jksmith      Eqw   05/21/2010 14:14:05
4319839 0.50000 qpostPro   jksmith      Eqw   05/21/2010 14:14:11
4319841 0.50000 qpostPro   jksmith      Eqw   05/21/2010 14:14:17
4319843 0.50000 qpostPro   jksmith      Eqw   05/21/2010 14:14:23
```

grep: “Regular expressions”

phonebook

```
Jacob   902 555 1234
James   902 555 4567
Jane    902 555 2365
Keith   902 555 5673
Kenny   902 555 0989
Larry   902 555 6575
```

grep 'Jacob' phonebook

```
Jacob   902 555 1234
```

grep -v 'Jac' phonebook
...everything but Jac...

```
James   902 555 4567
Jane    902 555 2365
Keith   902 555 5673
Kenny   902 555 0989
Larry   902 555 6575
```

PATTERNS	MATCHED STRINGS
Ja.e	“Jame” or “Jane” or “Ja8e”, *not* “Jaco”
Ja[mn]e	“Jame” or “Jane” *only*
Ja[^mn]e	“Jake” or “Ja8e” but *not* “Jame” or “Jane”
ab*c	a[0 or more b]c, e.g. ac, abc, abbc, abbbc
ab+c	a[1 or more b]c, e.g. abc, abbc, *not* ac
^abc	abc only at the beginning of a line
abc$	abc only at the end of a line
a(bc|de)f	abcf *or* adef

Extracting columns: cut

cut -c1-5 phonebook →

```
Jacob
James
Jane
Keith
Kenny
Larry
```

cut -c1-6,16- phonebook →

```
Jacob 1234
James 4567
Jane  2365
Keith 5673
Kenny 0989
Larry 6575
```

Extracting columns: cut

cat temp1

```
4319837 0.50000 qpostPro   jksmith      Eqw   05/21/2010 14:14:05
4319839 0.50000 qpostPro   jksmith      Eqw   05/21/2010 14:14:11
4319841 0.50000 qpostPro   jksmith      Eqw   05/21/2010 14:14:17
4319843 0.50000 qpostPro   jksmith      Eqw   05/21/2010 14:14:23
```

cut -c1-8 temp1

```
4319837
4319839
4319841
4319843
```

To avoid counting characters, try 'awk'.

Stream editor sed

sed 's/902/ /' phonebook

substitution

```
Jacob     555 1234
James     555 4567
Jane      555 2365
Keith     555 5673
Kenny     555 0989
Larry     555 6575
```

sed '4,5d' phonebook

deletion

```
Jacob   902 555 1234
James   902 555 4567
Jane    902 555 2365
Larry   902 555 6575
```

sed 's/^/hello /' phonebook

sub at start of line

```
hello Jacob   902 555 1234
hello James   902 555 4567
hello Jane    902 555 2365
hello Keith   902 555 5673
hello Kenny   902 555 0989
hello Larry   902 555 6575
```

Back to the example...

Suppose you have fifty jobs on the cluster, some in *r* state, some in *qw* state and some in *Eqw* state

How do you delete your *Eqw* jobs efficiently?

```
qstat > temp1

grep 'Eqw' temp1 > temp2

cut -c1-8 temp2 > temp3

sed 's/^/qdel /' temp3 > del.sh

source del.sh

rm temp1 temp2 temp3 del.sh
```

Eqw.sh

Use the pipe!

Avoid all those annoying intermediate files with “the pipe”:

“Pipes” output of first command into input of second

pipeEqw.sh

```
qstat | grep Eqw | cut -c1-8 | sed 's/^/qdel /' > del.sh

source del.sh

rm del.sh
```

Programming: Arguments and variables

```
#!/bin/sh
user="$1"
if  who | grep "^$user " > /dev/null
then
     echo "$user is logged ON"
else
     echo "$user is not logged on"
fi
```

iftest.sh

- $1, $2, *etc.* are script arguments
- $user is a variable
  - set with “name=”
  - referenced with “$name”

Programming: if – then – else

If
then
else
fi

```
#!/bin/sh
user="$1"
if  who | grep "^$user " > /dev/null
then
     echo "$user is logged ON"
else
     echo "$user is not logged on"
fi
```

iftest.sh

- Obeys *return code* of conditional, not *output!*
  - Experiment with 'echo $?' to see return codes
  - “0” good→ “then” clause; “nonzero” bad→ “else” clause

Programming: if – then – else

If
then
else
fi

```
#!/bin/sh
user="$1"
if  who | grep "^$user " > /dev/null
then
     echo "$user is logged ON"
else
     echo "$user is not logged on"
fi
```

iftest.sh

- Dialect-dependent!
  - /bin/tcsh has different syntax
  - first line #! tells system which shell to use

File testing example

```
#!/bin/sh

if [ -f $1 ]
then
     echo "Yes, $1 is here"
else
     echo "No file named $1 here"
fi
```

iftest2.sh

Checking files is very common,
so a collection of file tests are supplied...

File tests

Operator		Returns TRUE (exit status of 0) if
-e *file*	--------->	*file exists*
-d *file*	--------->	*file is a directory*
-f *file*	--------->	*file is an ordinary file*
-r *file*	--------->	*file is readable by the process*
-w *file*	--------->	*file is writable by the process*
-x *file*	--------->	*file is executable*
-s *file*	--------->	*file has nonzero length*
-L *file*	--------->	*file is a symbolic link*

Another practical example

You want to submit a number of jobs, differing only in the Temperature in the input file and you want each to have a unique job name.

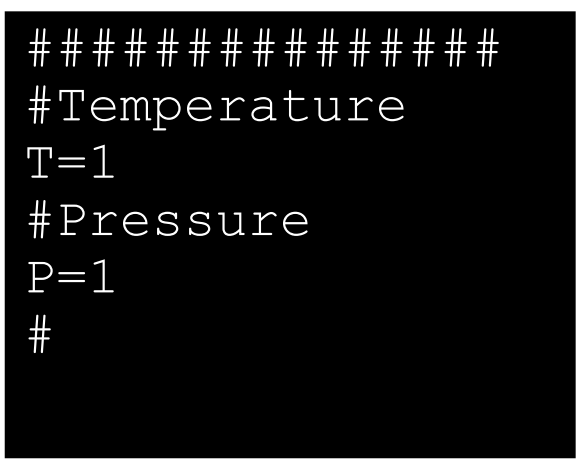
Another practical example

We'll want these files...

run.exe

template.in

sub.sh



```
################
#Temperature
T=1
#Pressure
P=1
#
```

```
#$ -cwd
#$ -N CHANGE
#$ -o t.out
#$ -e t.err
#$ -l h_rt=12:00:00

./run.exe input"$1".in
```

Programming: 'for' loop

make_input.sh

```
#!/bin/sh
# Make input files for a range of temperatures
j=0
for t in 178 189 208 213 218 228 248
do
   j=$((j+1))
   sed 's/T=1/T='$t'/'     template.in > input"$j".in
   sed 's/CHANGE/job'$j'/' sub.sh      > sub"$j".sh
   # 'echo' what we'd do instead of really submitting
   # the jobs, since this is just a demo:
   echo temperature=$t : qsub sub"$j".sh $j
done
```

Look into “here documents” for a way to do this without 'sed'.
Bash syntax shown, tcsh syntax is different.

Where next?

- Online manual, 'man'
  - **`man bash`** *or* **`man tcsh`**
- Tutorials on the Web
  - Google 'unix shell script tutorial'
- Books
  - *The Unix Programming Environment* by Kernighan & Pike is the classic
  - ...but there are many others!
- ACEnet Wiki
  - http://www.ace-net.ca/
- ACEnet Tech Support email
  - support@ace-net.ca

Example files

glooscap.ace-net.ca:/home/rdickson/demo/Scripting/*

cp /home/rdickson/demo/Scripting/* .